

$$SSE = \sum_{i=1}^n \left(\underset{\substack{\downarrow \\ y_{\text{actual}}}}{y_i} - \underset{\substack{\downarrow \\ y_{\text{predicted}}}}{h_{\theta}(x)} \right)^2$$

Cost function

$$J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(\underset{\substack{\downarrow \\ y_{\text{actual}}}}{y_i} - \underset{\substack{\downarrow \\ y_{\text{predicted}}}}{h_{\theta}(x)} \right)^2$$

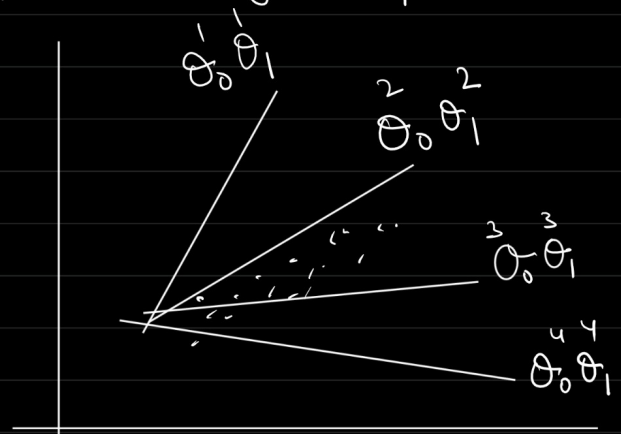
mean squared error

n = no of data points
 y = actual value
 $h_{\theta}(x)$ = predicted value

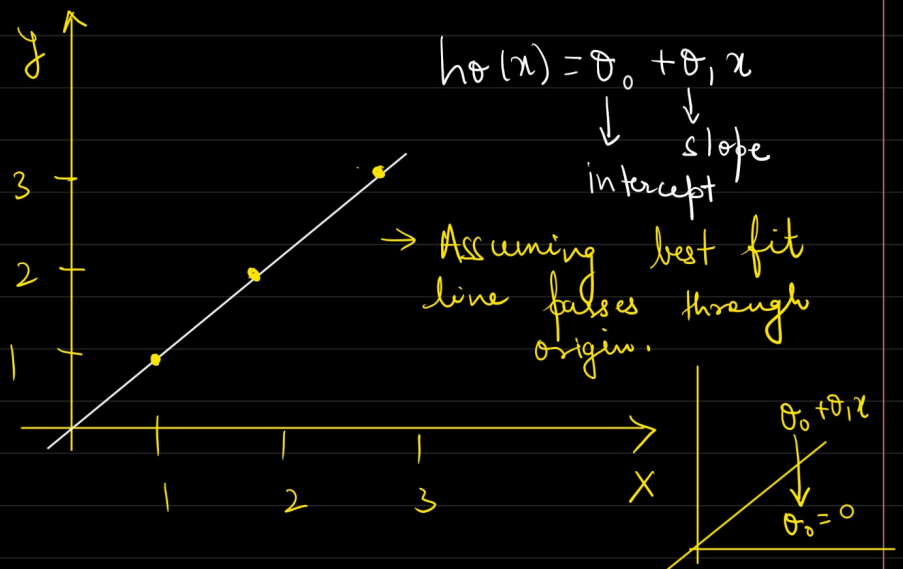
$$\begin{aligned} h_{\theta}(x) = y_{\text{pred}} &= \theta_0 + \theta_1 x \\ &= \beta_0 + \beta_1 x \\ &= c + mx \end{aligned}$$

$$\min J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - h_{\theta}(x_i))^2$$

* Aim : $\rightarrow$ minimise the cost function to get optimal coefficient

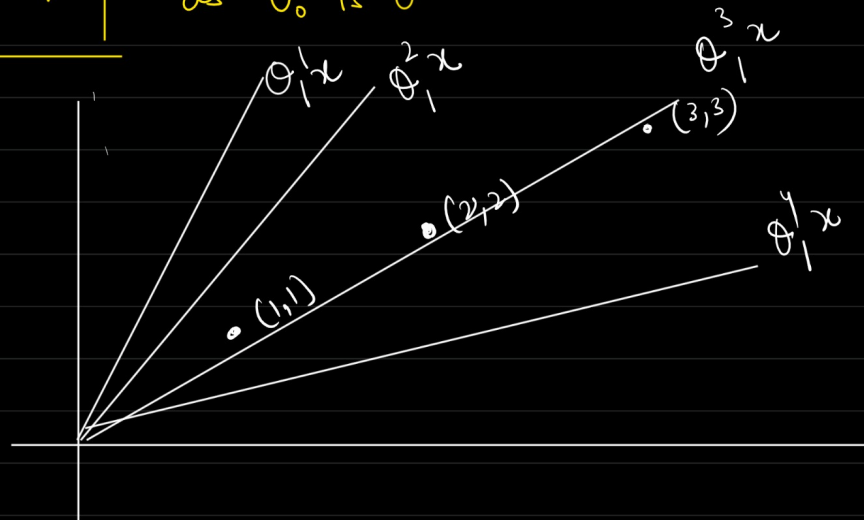


x	y
1	1
2	2
3	3



$$h_0(x) = \theta_1 x$$

as θ_0 is 0



Scen-1

$$h_0(x) = \theta_1 x \text{ as } \theta_0 = 0$$

assuming $\theta_1 = 1$

$$h_0(x) = \theta_1 x$$

$$x=1, \theta_1=1, \theta_0=0$$

$$h_0(x) = 1 \times 1 + 0$$

$$h_0(x) = y_{\text{pred}} = 1$$

$$x=2, h_0(x) = \theta_0 + \theta_1 x$$

$$= 0 + 1 \times 2$$

$$h_0(x) = y_{\text{pred}} = 2$$

$$x=3, h_0(x) = \theta_0 + \theta_1 x$$

$$= 0 + 1 \times 3$$

$$h_0(x) = 3$$

Scen2

$$\theta_1 = 0.5$$

$$x=1, h_0(x) = \theta_0 + \theta_1 x$$

$$= 0 + 0.5 \times 1$$

$$= 0.5$$

$$x=2 \Rightarrow 0 + 0.5 \times 2$$

$$\Rightarrow h_0(x) = 1$$

$$x=3 \Rightarrow 0 + 0.5 \times 3$$

$$h_0(x) = 1.5$$

$$\theta_0 = 0, \theta_1 = 0.5, h_0(x) = 0 + 0.5 \times x$$

x	y _{act}	y _{pred}
1	1	0.5
2	2	1
3	3	1.5

$$= J(\theta_1) = \frac{1}{3} [(1-0.5)^2 + (2-1)^2 + (3-1.5)^2]$$

Scen3

$$\theta_0 = 0, \theta_1 = 0$$

$$h_0(x) = 0 + 0 \times x$$

$$x=1 = 0$$

$$x=2 = 0$$

$$x=3 = 0$$

$$\theta_0 = 0, \theta_1 = 0$$

x	y	y _{pred}
1	1	0
2	2	0
3	3	0

$$J(\theta_1) =$$

$$\frac{1}{3} [(1-0)^2 + (2-0)^2 + (3-0)^2]$$

$$= \frac{1}{3} (1 + 4 + 9)$$

x	y_{act}	y_{pred}
1	1	1
2	2	2
3	3	3

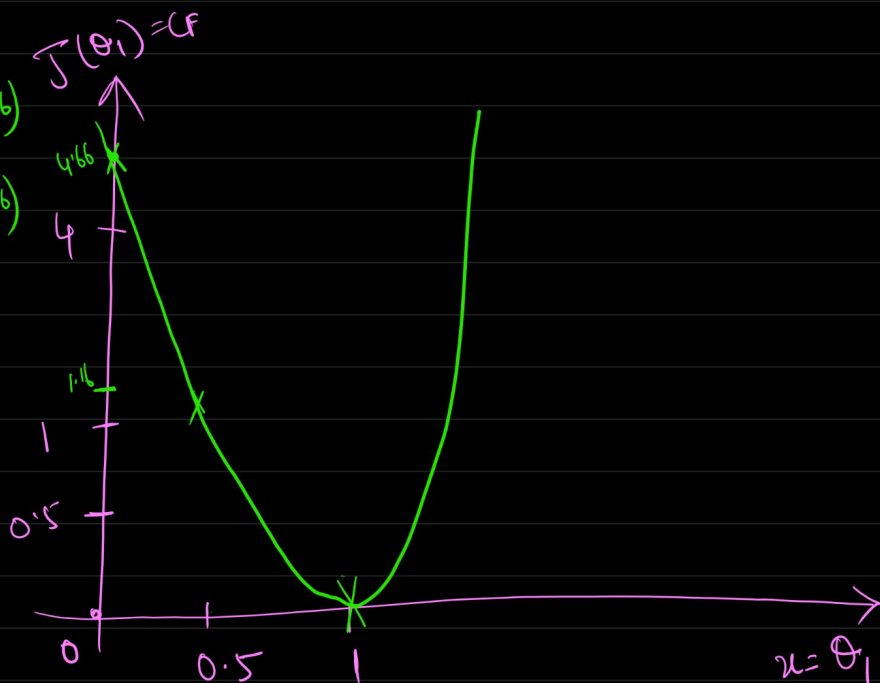
$$= \frac{1}{3} [(0.25 + 1 + 2.25)] = \frac{14}{3} = \underline{\underline{4.66}}$$

$$= \underline{\underline{1.16}}$$

$$J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - h_{\theta}(x_i))^2$$

$$J(\theta_1) = \frac{1}{3} [(1-1)^2 + (2-2)^2 + (3-3)^2] = 0$$

$$\theta_0 = 0 \begin{cases} \theta_1 = 1, J(\theta_1) = 0 \quad (1, 0) \\ \theta_1 = 0.5, J(\theta_1) = 1.16 \quad (0.5, 1.16) \\ \theta_1 = 0, J(\theta_1) = 4.66 \quad (0, 4.66) \end{cases}$$

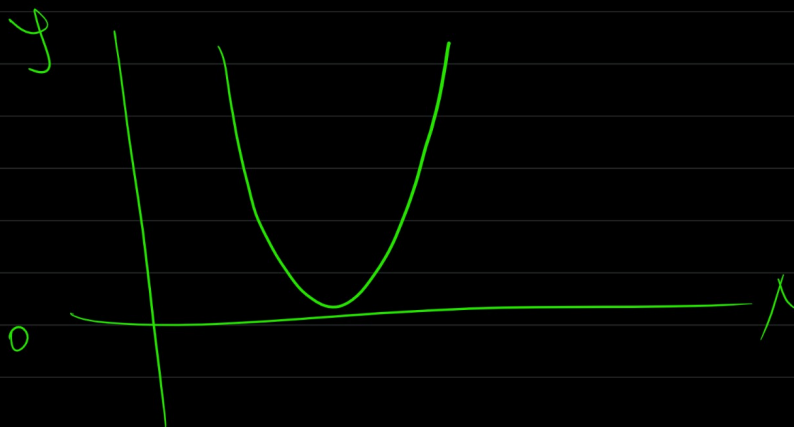


$$J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - h_{\theta}(x_i))^2$$

min

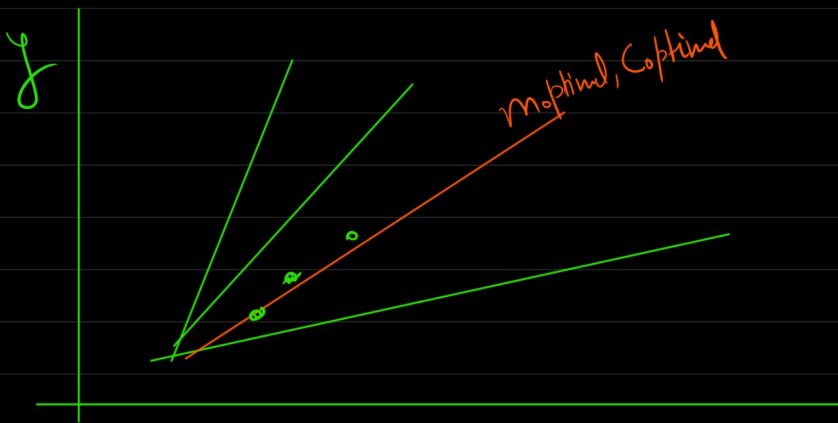
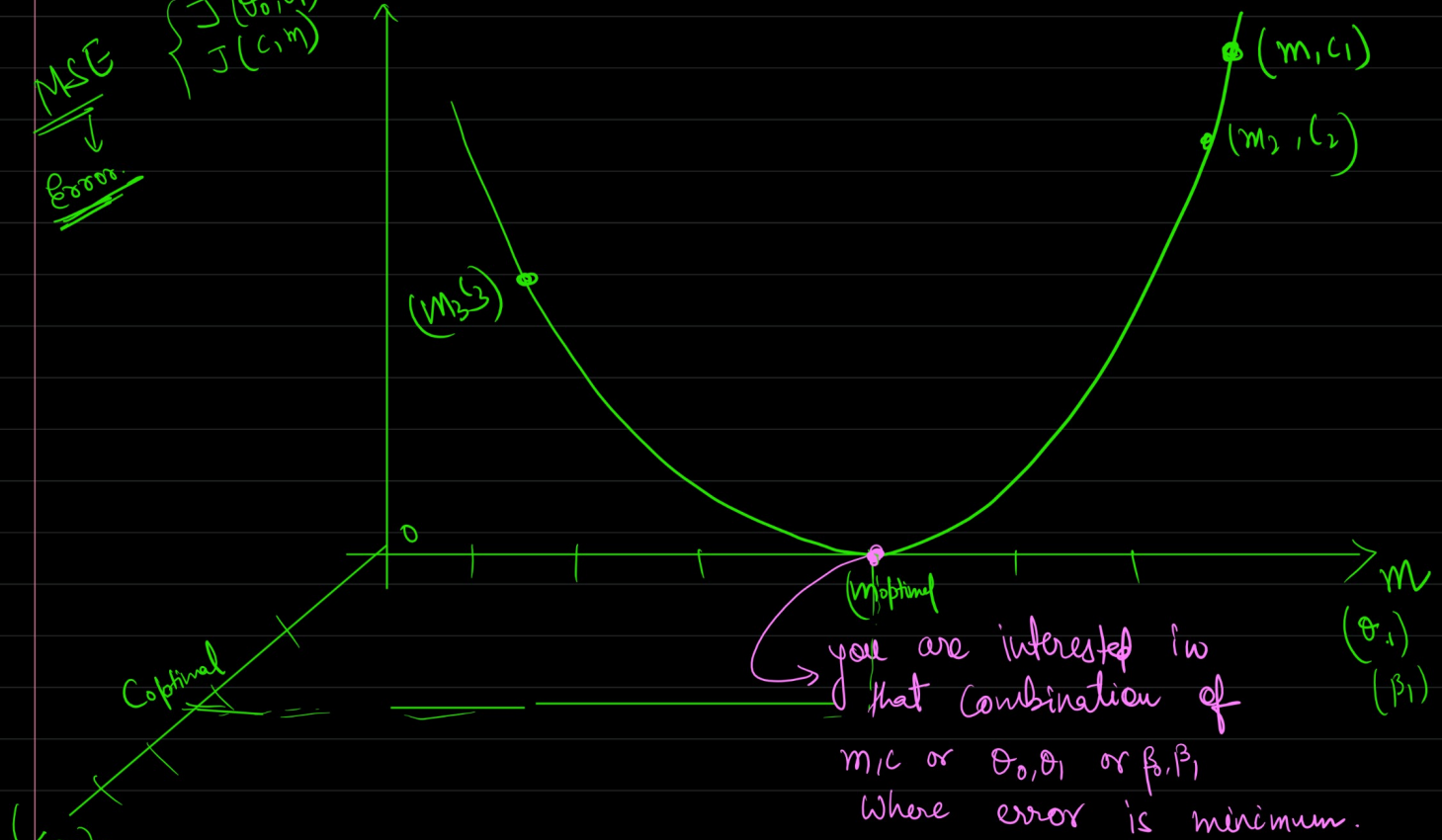
$y = x^2$

Parabola



MSE
↓
Error

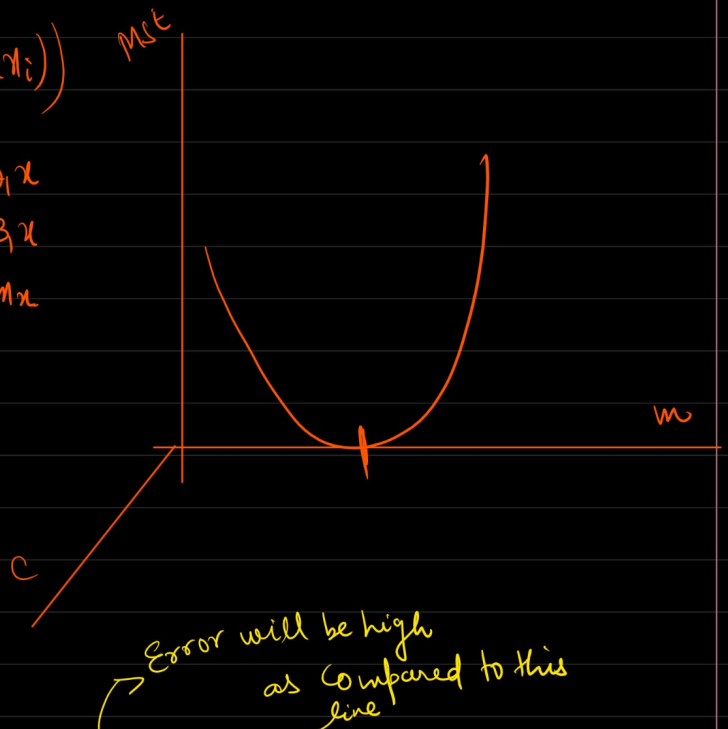
$\begin{cases} J(\theta_0, \theta_1) \\ J(c, m) \end{cases}$

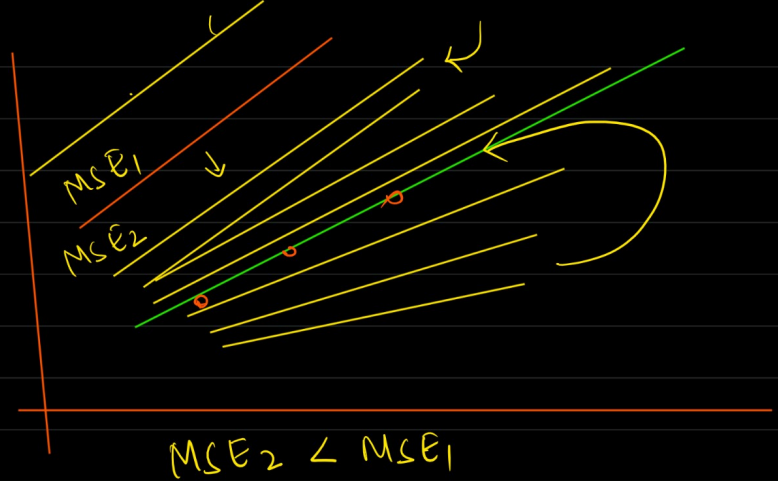


$$J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - h_{\theta}(x_i))^2$$

$\theta_0 + \theta_1 x$
 $\beta_0 + \beta_1 x$
 $c + mx$

→ infinite possibilities of m, c .





* Convergence Algorithm

→ keep moving new best fit line in the direction of the datapoints, until error (MSE/CF) is reduced as compared to previous lines.

Convergence Algorithm (Called as Gradient Descent)

Repeat until convergence → learning rate (eta)

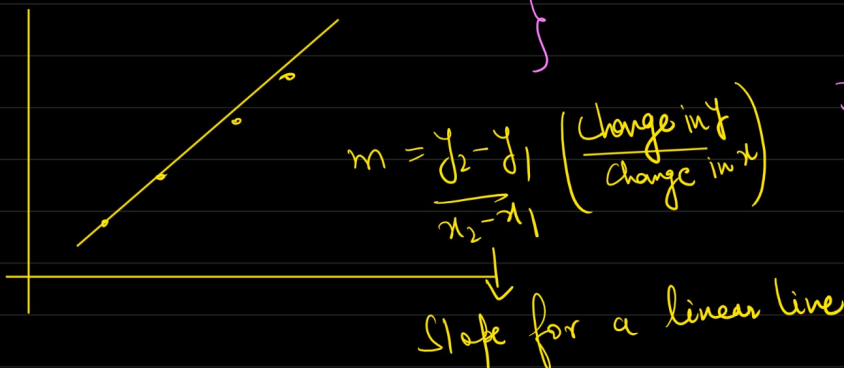
$$\begin{cases} m_{\text{new}} = m_{\text{old}} - \eta \frac{\partial (CF)}{\partial m_{\text{old}}} \\ c_{\text{new}} = c_{\text{old}} - \eta \frac{\partial (CF)}{\partial c_{\text{old}}} \end{cases}$$

Repeat until convergence (simultaneously)

$$\theta_j : \theta_j - \eta \frac{\partial J(\theta_j)}{\partial \theta_j}$$

$\theta_1 = \theta_1 - \eta \frac{\partial (CF)}{\partial \theta_1} \rightarrow m_{\text{new}} \text{ old } m_{\text{old}}$
 $\theta_0 = \theta_0 - \eta \frac{\partial (CF)}{\partial \theta_0} \rightarrow c_{\text{new}} \text{ old } c_{\text{old}}$

$J = 0.1$





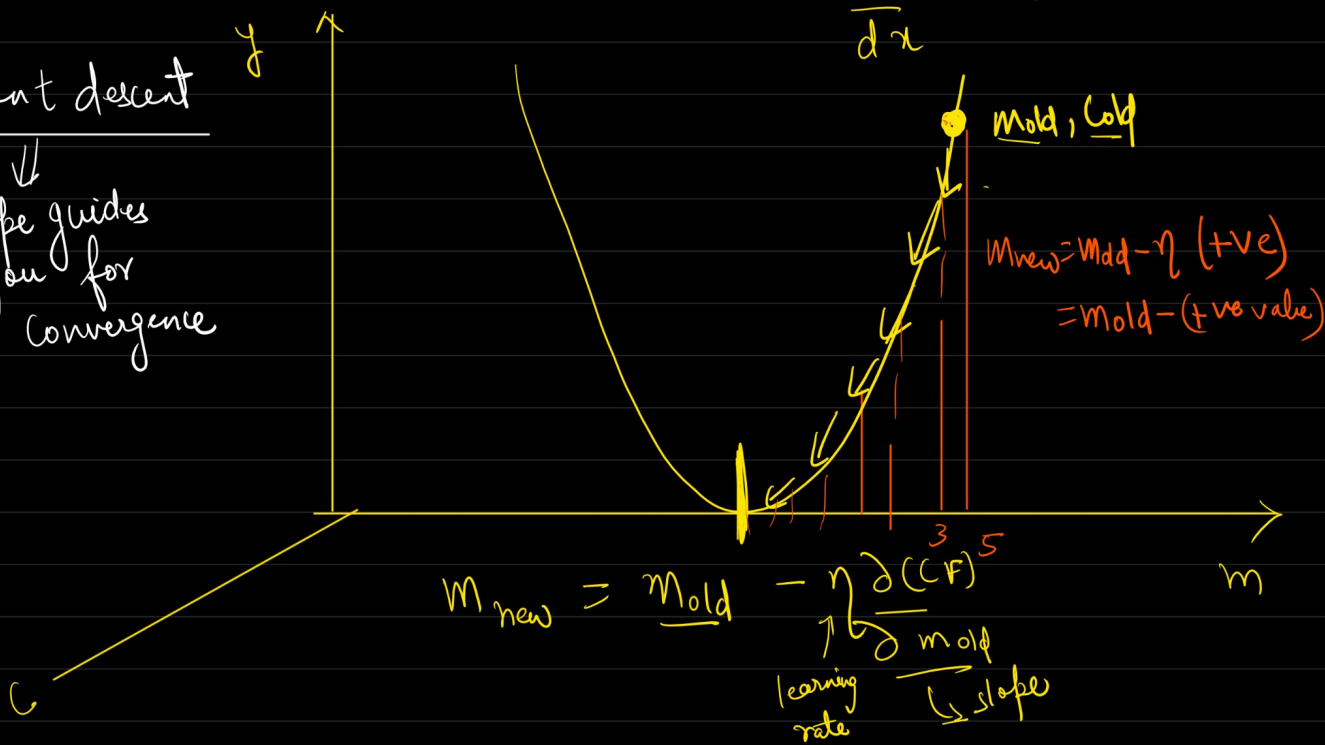
derivative $\downarrow$ $\frac{dy}{dx} \approx \frac{\partial y}{\partial x}$

partial derivative $\downarrow$ $\frac{\partial y}{\partial x}$

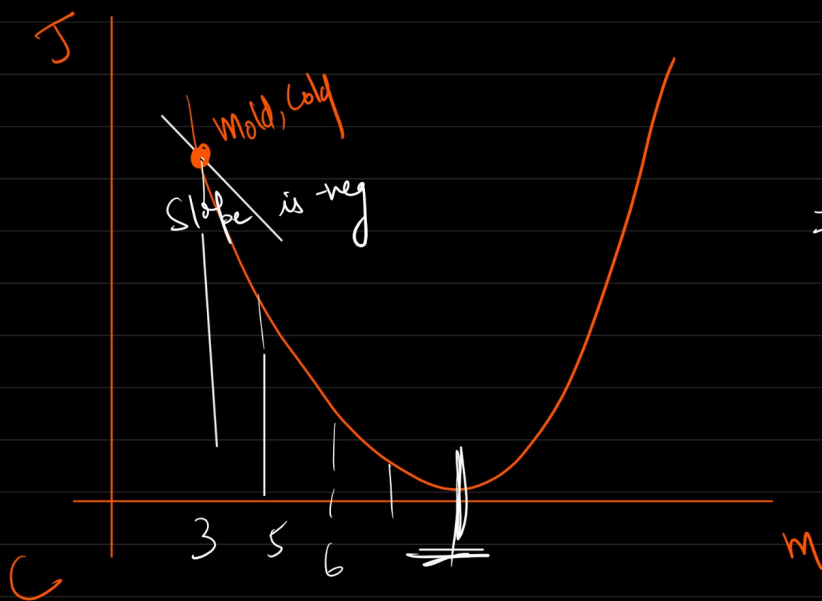
$$\frac{dx^n}{dx} = \eta x^{n-1}$$

Gradient descent

$\downarrow$
Slope guides you for convergence



$$C_{new} = C_{old} - \eta \frac{\partial (CF)}{\partial C_{old}}$$



Note about the slope

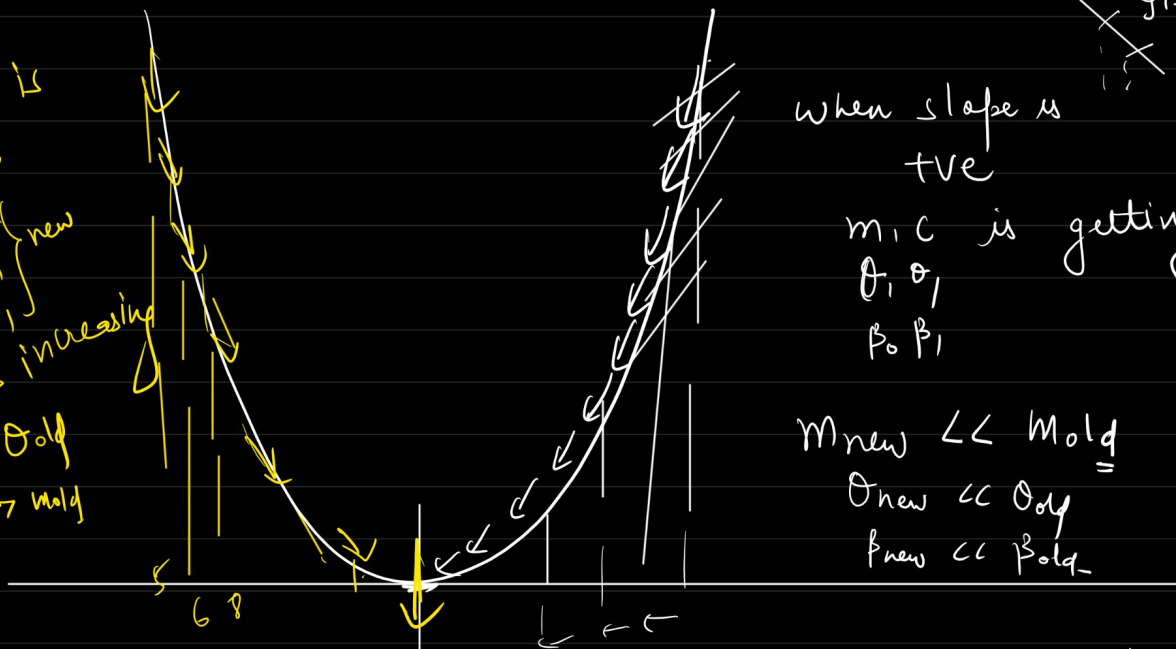


$$\frac{y_2 - y_1}{x_2 - x_1} \geq +ve$$

When slope is -ve

m, c, θ, β new
is increasing

$\theta_{new} > \theta_{old}$
 $m_{new} > m_{old}$



When slope is +ve

m, c is getting reduced
 θ, β

$m_{new} < m_{old}$
 $\theta_{new} < \theta_{old}$
 $\beta_{new} < \beta_{old}$

~~y is decreasing~~
~~slope is -ve~~

$$m_{new} = m_{old} - \eta \frac{\partial (CF)}{\partial m_{old}}$$

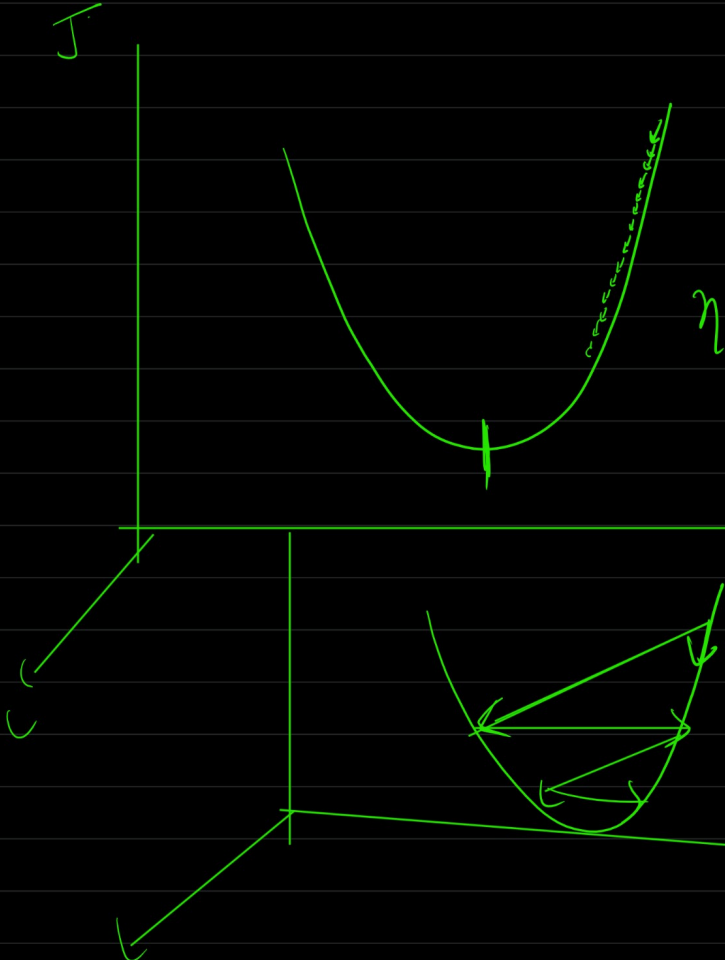
$$c_{new} = c_{old} - \eta \frac{\partial (CF)}{\partial c_{old}}$$

slope η
 $\downarrow \quad \uparrow$
 5×0.02
will be very less than 5

$\eta \rightarrow$ learning rate

It decides the convergence speed.

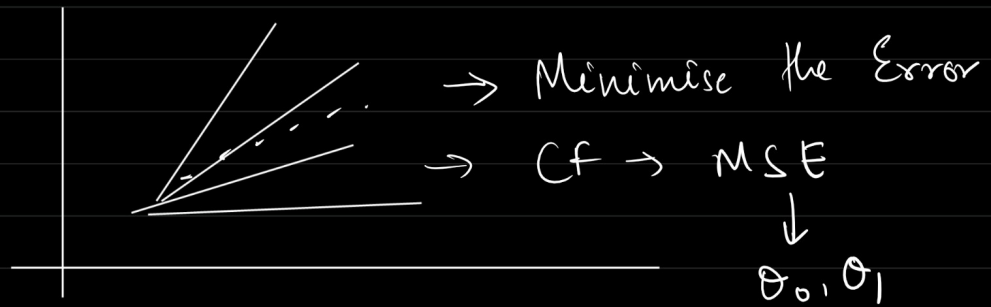
η - very very small
A lot of time will take for convergence



η - very big no then it overshoots the minima
 $m_{new} = m_{old} - \eta$ (+ve)

$$\eta = 0 \text{ to } 1$$

Conclusion



Gradient descent → Repeat until convergence

$$\theta_j : \theta_j - \eta \frac{\partial J(\theta_j)}{\partial \theta_j}$$

